

The Role of AI in Advancing Scientific Research from Hypothesis to Publication

Ameera F. Mohammad*

Engineering Requirements Unit, United
Arab Emirates University
Abu Dhabi, Al Ain, United Arab
Emirates
a.fares@uaeu.ac.ae

Waleed Ahmed*

Engineering Requirements Unit, United
Arab Emirates University
Abu Dhabi, Al Ain, United Arab
Emirates
w.ahmed@uaeu.ac.ae

Abdurrahman G. Almekhlafi

Department of Education, College of
Arts, Humanities and Social Sciences
University of Sharjah
Sharjah, United Arab Emirates
<https://orcid.org/0000-0002-8988-0803>

Salem Alzahmi

Chemical and Petroleum Engineering
Department, College of Engineering,
United Arab Emirates University
Abu Dhabi, Al Ain, United Arab
Emirates
s.alzahmi@uaeu.ac.ae

Ali H. Al-Marzouqi

Chemical and Petroleum Engineering
Department, College of Engineering,
United Arab Emirates University
Abu Dhabi, Al Ain, United Arab
Emirates
hassana@uaeu.ac.ae

Saed Alrabae

Information Systems & Security-
Collage of Information Technology,
United Arab Emirates University
Abu Dhabi, Al Ain, United Arab
Emirates
salrabae@uaeu.ac.ae

Essam Zaneldin

Civil & Environmental Engineering
Department, College of Engineering,
United Arab Emirates University
Abu Dhabi, Al Ain, United Arab
Emirates
essamz@uaeu.ac.ae

Abstract Artificial Intelligence (AI) has rapidly evolved from a supportive computational tool to a transformative driver of scientific research, permeating nearly every stage of the research lifecycle. No longer confined to isolated applications, AI is now integrated into the broader ecosystem of inquiry, reshaping the way hypotheses are formulated, data are collected and analyzed, experiments are designed, and findings are disseminated. Breakthroughs in large language models (LLMs), retrieval-augmented generation (RAG), multimodal AI, and agentic workflows have expanded the scope of research capabilities, enabling systems that not only process information but also reason across disciplines, generate novel insights, and automate complex workflows. This paper presents a comprehensive narrative review of 36 recent studies that examine AI's contributions to scientific research, with particular attention to its integration across key phases: literature discovery, hypothesis generation, experimental design, data analysis, visualization, manuscript drafting, and peer review. In addition, the review highlights AI's applications in STEM, biomedical sciences, social sciences, and interdisciplinary domains, demonstrating its potential to accelerate innovation and bridge methodological gaps. The role of AI in systematic reviews and meta-analyses is emphasized, underscoring its ability to improve reproducibility and efficiency while reducing research timelines. Beyond opportunities, the review critically examines risks including bias, hallucination, overfitting, data privacy, authorship, and academic integrity. Ethical frameworks such as model cards, AI usage cards, and international governance standards are explored as essential mechanisms for ensuring transparency, fairness, and accountability. The findings suggest that AI should be regarded not as a substitute for human inquiry but as a collaborative partner that amplifies creativity, enhances rigor, and accelerates the production of scientific knowledge. By embedding ethical safeguards and reproducibility standards, AI can serve as both a catalyst for discovery and a guardian of research integrity in the evolving landscape of twenty-first century science.

Keywords—Artificial Intelligence, Large Language Models, Research Lifecycle, Scientific Discovery, Reproducibility, Research Integrity, Governance, and Ethics.

I. INTRODUCTION

The rise of artificial intelligence represents one of the most significant paradigm shifts in the conduct of science since the introduction of digital computing. While traditional computational tools were primarily designed to automate repetitive or routine data processing tasks, modern AI systems extend far beyond automation by incorporating adaptive learning, contextual reasoning, and pattern recognition at scales that surpass human capacity. AI is increasingly capable of simulating reasoning processes, synthesizing knowledge from vast and heterogeneous datasets, and generating insights that both parallel and extend human cognition. This capability allows researchers not only to accelerate established workflows but also to pursue entirely new forms of inquiry that were previously impractical or unattainable. Its applications now span the entire research lifecycle: from literature discovery and hypothesis generation to experimental design, data collection and analysis, simulation, and visualization, culminating in manuscript drafting, peer review, and publication. In each of these stages, AI introduces opportunities for enhanced efficiency, reproducibility, and interdisciplinary integration, while simultaneously raising important questions regarding reliability, transparency, and governance [1-6]. Large language models (LLMs) have particularly accelerated this transformation. Their ability to generate context-aware text, summarize vast corpora, and provide conversational interfaces has made them integral in both education and research [7]. Retrieval-augmented generation (RAG) frameworks further extend these

capabilities by connecting LLMs to external databases, reducing hallucinations, and ensuring knowledge remains current [8]. More recently, agentic workflows and multi-agent RAG systems have introduced autonomous orchestration of research tasks, suggesting a future where AI agents perform literature reviews, design experiments, and validate results collaboratively with humans [9]. At the same time, challenges persist. Scholars such as Gundersen and Kjensmo [3] have warned of widespread reproducibility failures in AI systems, while Hosseini and Resnik [10] noted ethical risks associated with AI-assisted misconduct detection. Concerns about bias [11], privacy [12], and authorship [13] underscore the need for governance frameworks that can strike a balance between innovation and accountability.

We aim to provide a comprehensive narrative review of AI's expanding role across the research lifecycle, sitting it within the broader context of contemporary scientific practice. We synthesize findings from numerous key studies, drawing on diverse perspectives from computer science, engineering, biomedical research, and the social sciences. The discussion is organized into sections that address lifecycle integration, emerging AI approaches, disciplinary applications, evaluation metrics, case studies, systematic reviews, risks and limitations, research integrity, governance, and future directions. By systematically examining these dimensions, the paper highlights both the opportunities and challenges inherent in embedding AI into scientific workflows. Importantly, we argue that AI should not be viewed merely as a technological tool or as a substitute for human inquiry, but rather as an active collaborator—one that amplifies creativity, accelerates discovery, and strengthens the rigor, reproducibility, and interdisciplinary scope of modern scientific practice.

II. AI ACROSS THE RESEARCH LIFECYCLE

This paper adopts a conceptual framing in which artificial intelligence functions as a multi-dimensional enabler across the research lifecycle supporting cognitive augmentation in knowledge discovery and hypothesis generation, computational acceleration in data analysis and modeling, and integrity reinforcement through peer review and reproducibility.

A. Literature Discovery

AI has revolutionized the initial stage of the research lifecycle, literature discovery. Zhou et al. [14] demonstrated that AI research support systems can reduce the time required to identify relevant studies by more than half, while also ensuring broader coverage of the literature. Exitance [15] observed that as scientific publications continue to grow exponentially, AI systems act as filters that transform overwhelming information into structured, actionable knowledge. The integration of AI across the research lifecycle can be viewed through three complementary functions: cognitive augmentation (literature discovery, hypothesis generation), computational acceleration (data analysis, simulation, modeling), and integrity reinforcement (manuscript drafting, peer review, reproducibility). This triadic framework underpins the synthesis of findings presented in subsequent sections.

B. Hypothesis Generation

AI also plays a crucial role in hypothesis generation. Singh et al. [16] introduced the concept of agentic RAG, where

autonomous AI agents generate and test hypotheses by integrating multiple data streams. Gao et al. [17] argued that advanced RAG architectures enable contextualized reasoning, allowing hypotheses to emerge from cross-disciplinary linkages that human researchers may overlook.

C. Experimental Design

In experimental design, Wang et al. [4] highlighted how self-supervised learning and geometric deep learning have improved the accuracy of simulation models, leading to more efficient hypothesis testing. Ajayi [18] emphasized that AI-driven methodologies enable researchers in STEM to explore parameter spaces previously inaccessible due to computational constraints.

D. Data Collection and Analysis

Data collection increasingly involves AI-enhanced automation. Chávez-Ángel et al. [19] showed how AI improves spectroscopy and microscopy, enabling more efficient pattern recognition in complex data. Zuccarini et al. [20] demonstrated how machine learning accelerates modeling in materials science, reducing both cost and time.

E. Visualization and Simulation

Visualization is essential for interpreting results. Badini et al. [2] reported that AI-generated visual models facilitate understanding of high-dimensional data. In climate modeling and biomedical research, AI-based simulations reveal complex system interactions, producing insights not possible with traditional statistical methods [21].

F. Manuscript Drafting and Peer Review

Finally, AI assists with manuscript preparation and peer review. Checco et al. [22] demonstrated how AI can predict peer-review outcomes and match reviewers to manuscripts, while Fabiano et al. [23] showed that AI reduces the time required for systematic reviews by up to 52%. Hosseini and Resnik [10] warned, however, that AI-assisted misconduct detection must be implemented with caution to avoid false positives and ethical breaches.

Figure 1 illustrates the integration of AI across research stages, while Table I summarizes its roles and benefits throughout the lifecycle.

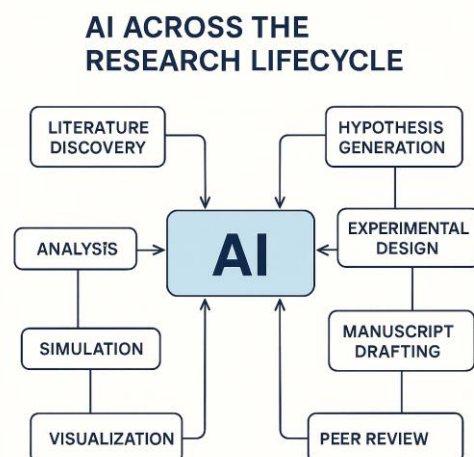


Fig. 1. Integration of Artificial Intelligence in Scientific Research Stages

In fact, Figure 1 is extended here as a conceptual framework positioning AI as an integrative catalyst across three interrelated research functions: cognitive augmentation (e.g., literature discovery, hypothesis generation), computational acceleration (e.g., experimental design, modeling), and integrity reinforcement (e.g., peer review, reproducibility). This taxonomy underpins the synthesis presented throughout this paper.

TABLE I. ROLE OF AI ACROSS RESEARCH STAGES

Research Stage	AI Contributions	Key References
Literature Discovery	Automated searches, summarization, knowledge synthesis	[24, 25]
Hypothesis Generation	Cross-disciplinary reasoning, RAG systems, agentic AI	[17, 26]
Experimental Design	Simulation models, parameter optimization	[27, 28]
Data Collection & Analysis	High-throughput imaging, spectroscopy, pattern recognition	[29, 30]
Simulation & Visualization	Modeling complex datasets, predictive visualization	[31, 32]
Manuscript Drafting	Automated writing assistance, citation management	[33, 34]
Peer Review	Reviewer matching, outcome prediction, misconduct detection	[14, 35]

III. TYPES OF AI APPROACHES

The methods used to embed AI into research are diverse. LLMs such as GPT-like systems provide language fluency and knowledge integration, but are vulnerable to hallucinations [36]. RAG frameworks extend these models by connecting them to external databases, ensuring that outputs remain grounded in factual information [37]. Singh et al. [16] expanded on this by introducing agentic RAG, in which multiple autonomous agents manage tasks dynamically. Salve et al. [38] further described multi-agent RAG systems, where specialized agents handle heterogeneous data sources, improving scalability and accuracy.

AutoML frameworks simplify model development by automating algorithm selection and hyperparameter optimization, making advanced AI methods accessible to researchers without deep expertise [39]. Multimodal AI, which integrates text, image, and sensor data, has been shown to enhance biomedical and climate modeling [40]. Finally, code generation tools embedded in LLMs have proven helpful for simulation and algorithm development, though concerns remain about replicating insecure coding practices [41].

Figure 2 provides an overview of these approaches and their interactions.

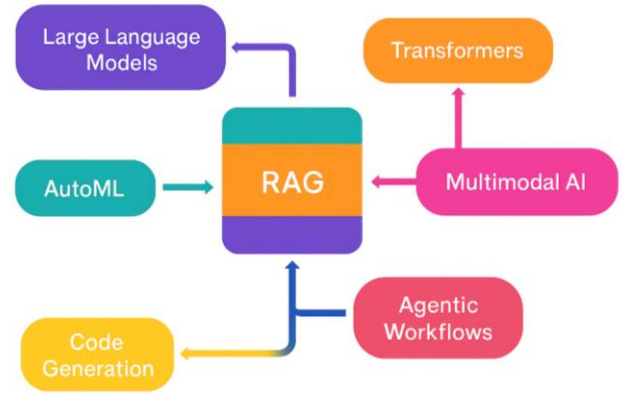


Fig. 2. Conceptual Illustration of Artificial Intelligence as an Integrated Research Enabler.

IV. APPLICATIONS ACROSS DISCIPLINES

The applications of AI span a wide range of domains.

In STEM, AI enhances data acquisition, optimizes experimental design, and accelerates modeling. Zuccarini et al. [20] demonstrated its role in materials discovery, while Badini et al. [2] and Chávez-Ángel et al. [19] showed breakthroughs in materials design and microscopy.

In biomedical sciences, AI contributes to genomics, molecular simulation, and drug discovery pipelines. Wang et al. [4] and Chávez-Ángel et al. [19] noted that AI supports the design of novel compounds and analysis of genomic sequences, improving efficiency and accuracy.

In the social sciences, AI applications extend to economics, psychology, and marketing. Alias et al. [42] reported that AI tools improve forecasting and decision-making, while Sanieva [7] highlighted AI's role in facilitating interdisciplinary research methods.

Kusters et al. [43] argued that interdisciplinary research is particularly enabled by AI, which bridges traditional boundaries and fosters synergies across diverse fields.

Figure 3 illustrates AI-driven innovations across STEM, biomedical, social sciences, and interdisciplinary domains.

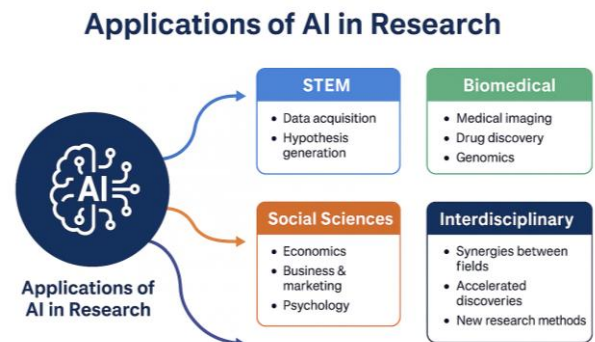


Fig. 3. AI-Driven Innovations in STEM, Biomedical, Social Sciences, and Interdisciplinary Research

V. EVALUATION AND METRICS

Assessing the reliability of AI in research requires careful evaluation. Calster et al. [44] emphasized the use of discrimination metrics such as AUROC, while Handelman et al. [45] recommended decision-analytic measures such as calibration for clinical applications. Gundersen and Kjensmo [3] observed that reproducibility remains a challenge, with many studies failing to document essential variables. Mortaji and Sadeghi [46] proposed robustness and transparency metrics to address vulnerabilities, including adversarial attacks and data quality issues.

Table II summarizes evaluation metrics for AI systems, highlighting their relevance for reproducibility, interpretability, and scalability.

TABLE II. EVALUATION METRICS FOR AI SYSTEMS IN RESEARCH

Metric	Description	Relevance	Key References
Accuracy / Precision	Proportion of correctly predicted outcomes vs. true outcomes	Ensures baseline model reliability	[45, 46]
AUROC / Discrimination	Measures the model's ability to distinguish between classes	Critical for reproducibility in biomedical AI	[45, 46]
Calibration	Agreement between predicted probabilities and observed outcomes	Improves interpretability and trust	[45]
Robustness	Model stability under noisy, incomplete, or adversarial inputs	Essential for scalability in real-world systems	[46]
Transparency & Explainability	Clarity of model decisions and feature contributions	Enhances interpretability, prevents bias	[1, 46, 47]
Reproducibility	Extent to which results can be replicated across settings and datasets	Core requirement for scientific rigor	[3, 23]
Time / Cost Efficiency	Reduction in resources required for data processing, analysis, or reviews	Promotes scalability and wider adoption	[23, 48]
Inter-rater Reliability	Consistency of results across multiple evaluators or reviewers	Strengthens reproducibility in peer review	[10, 22]

VI. CASE STUDIES

Several case studies illustrate AI's impact across disciplines. In materials science, Zuccarini et al. [20] described accelerated modeling of alloys, while Chávez-Ángel et al. [19] emphasized AI's role in spectroscopy and microscopy. In biomedical research, AI supports drug discovery pipelines by predicting interactions between molecules [49]. In climate science, AI improves predictive accuracy by integrating multimodal datasets, while in social sciences, Ajayi [18] highlighted AI's role in economic forecasting and behavioral modeling.

VII. SYSTEMATIC REVIEWS AND META-ANALYSES

AI is increasingly utilized in systematic reviews, where notable efficiency gains are observed. Fabiano et al. [23] and Guerrero-Solís et al. [48] reported reductions of up to 52% in review times, alongside improved screening accuracy. Tomczyk et al. [50] emphasized that human-AI collaboration remains essential for preserving interpretive rigor, ensuring that automation does not replace critical judgment.

VIII. RISKS AND LIMITATIONS

The integration of AI into research introduces risks. Hallucinations and biased outputs remain critical concerns for LLMs [51]. Privacy and data leakage risks have been highlighted in biomedical applications [52]. Overfitting and underfitting undermine trust in model predictions, while adversarial attacks raise vulnerabilities in sensitive domains. Haque et al. [53] concluded that LLMs may replicate insecure coding practices, necessitating robust guardrails.

Figure 4 summarizes these risks, categorizing them into hallucinations, bias, data vulnerabilities, and overreliance.

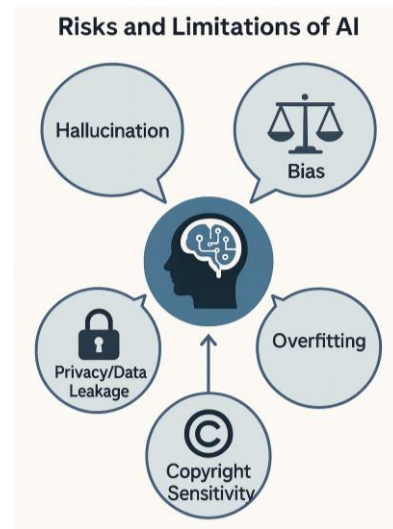


Fig. 4. Key Risks of AI: Bias, Hallucination, and Data Vulnerabilities

IX. RESEARCH INTEGRITY AND GOVERNANCE

AI also raises concerns for research integrity. Chen et al. [47] and Limongi [1] stressed the importance of transparency and ethical oversight. Hosseini and Resnik [10] noted risks in automated misconduct detection, while Leong and Clarke [54] discussed challenges in plagiarism and authorship detection. Governance frameworks, such as AI usage cards [55] and model cards [56] provide mechanisms for accountability. Ganesh et al. [55] and Brajovic et al. [57] extended these frameworks to institutional and international contexts, highlighting the importance of COPE, UNESCO, and OECD guidelines.

X. FUTURE DIRECTIONS

Future directions for AI in research point toward agentic AI systems, human-in-the-loop safeguards, and platforms for reproducibility. Gridach et al. [58] and Dawid et al. [59] highlighted agentic workflows as transformative for scientific discovery, while Satav [60] proposed enterprise API strategies for integration. Hughes et al. [61] emphasized the importance of energy-efficient AI deployment as a critical concern.

Integration with lab automation and Internet of Things (IoT) systems represents another important frontier, enabling AI-driven experimentation that seamlessly links data collection, real-time monitoring, large-scale analysis, and the rapid dissemination of results. When combined with autonomous laboratory equipment and sensor-rich environments, AI can orchestrate entire experimental pipelines, from hypothesis validation to high-throughput screening, thereby reducing human error and significantly accelerating discovery. Such interconnected systems hold the potential to establish “self-driving laboratories,” where continuous feedback loops between data acquisition, model refinement, and experimental adjustment enable unprecedented efficiency. Researchers should document and disclose AI tools, prompts, and datasets used to ensure transparency and reproducibility. Journals and peer reviewers are encouraged to adopt AI disclosure statements, reproducibility checklists, and ethics review protocols. Institutions should provide structured training and governance frameworks that promote responsible innovation while safeguarding research integrity.

XI. CONCLUSION

AI is fundamentally transforming the research enterprise, permeating nearly every stage of the scientific process, from hypothesis generation and experimental design to data analysis, visualization, peer review, and final publication. Its applications span across STEM disciplines, biomedical sciences, and the social sciences, where AI tools accelerate discovery, optimize experiments, enhance reproducibility, and create new opportunities for interdisciplinary collaboration. By reducing time-intensive tasks and providing advanced analytical capabilities, AI enables researchers to explore questions of greater complexity and societal relevance.

However, unresolved challenges remain significant. Issues such as algorithmic bias, hallucinations in large language models, data privacy and security risks, and concerns over authorship and academic integrity threaten to undermine the credibility of AI-assisted research. Addressing these challenges requires a multi-pronged approach that combines technical innovation with robust governance. Frameworks such as model cards and AI usage cards provide concrete pathways toward transparency and accountability, while emerging paradigms such as agentic AI point toward a future of autonomous yet ethically constrained workflows.

The balance between innovation and responsibility will ultimately determine AI’s long-term role in science. If integrated without safeguards, AI risks amplifying existing inequities, reinforcing opacity, and eroding public trust in research. Conversely, if governed responsibly through international standards, reproducibility protocols, and human-in-the-loop oversight, AI can serve not as a replacement for human inquiry but as a true collaborator. In this role, AI will amplify creativity, accelerate discovery, and safeguard the rigor, reproducibility, and ethical foundations of scientific knowledge production, ensuring its alignment with societal values and the enduring principles of responsible innovation.

ACKNOWLEDGMENT

The authors gratefully acknowledge the support of the United Arab Emirates University (UAEU) for providing the research environment, resources, and continuous encouragement that enabled the successful completion of this work. The collaborative and innovative atmosphere at UAEU

has been instrumental in advancing the study of artificial intelligence and its applications in scientific research.

REFERENCES

- [1] R. J. F. S. R. J. T. Limongi and Strategies, "The use of artificial intelligence in scientific research with integrity and ethics," vol. 16, no. 1, pp. e845-e845, 2024.
- [2] S. Badini, S. Regondi, and R. J. M. Pugliese, "Unleashing the power of artificial intelligence in materials design," vol. 16, no. 17, p. 5927, 2023.
- [3] O. E. Gundersen and S. Kjensmo, "State of the art: Reproducibility in artificial intelligence," in *Proceedings of the AAAI conference on artificial intelligence*, 2018, vol. 32, no. 1.
- [4] H. Wang *et al.*, "Scientific discovery in the age of artificial intelligence," vol. 620, pp. 47-60, 2023.
- [5] S. Li and Y. Wang, "Research on interdisciplinary characteristics: A case study in the field of artificial intelligence," in *IOP conference series: materials science and engineering*, 2019, vol. 677, no. 5, p. 052023: IOP Publishing.
- [6] Z. Chen *et al.*, "Research integrity in the era of artificial intelligence: Challenges and responses," vol. 103, no. 27, p. e38811, 2024.
- [7] A. D. J. E. I. U. P. Sanieva, RESHENIYA, "DEVELOPMENT OF INTERDISCIPLINARY SCIENTIFIC RESEARCH THROUGH THE APPLICATION OF ARTIFICIAL INTELLIGENCE," 2024.
- [8] Y. Ding *et al.*, "A Survey on RAG Meets LLMs: Towards Retrieval-Augmented Large Language Models," vol. abs/2405.06211, 2024.
- [9] A. Singh, A. Ehtesham, S. Kumar, and T. T. J. a. p. a. Khoei, "Agentic retrieval-augmented generation: A survey on agentic rag," 2025.
- [10] D. B. Resnik, M. J. A. Hosseini, and Ethics, "The ethics of using artificial intelligence in scientific research: new guidance needed for a new tool," vol. 5, no. 2, pp. 1499-1521, 2025.
- [11] J. W. Gichoya *et al.*, "AI pitfalls and what not to do: mitigating bias in AI," vol. 96, no. 1150, p. 20230023, 2023.
- [12] D. Elliott and E. J. F. i. A. I. Soifer, "AI technologies, privacy, and security," vol. 5, p. 826737, 2022.
- [13] R. M. Ballardini, K. He, and T. Roos, "AI-generated content: authorship and inventorship in the age of artificial intelligence," in *Online Distribution of Content in the EU*: Edward Elgar Publishing, 2019, pp. 117-135.
- [14] Z. Zhou *et al.*, "From Hypothesis to Publication: A Comprehensive Survey of AI-Driven Research Support Systems," vol. abs/2503.01424, 2025.
- [15] A. J. N. Extance, "How AI technology can tame the scientific literature," vol. 561, no. 7722, pp. 273-275, 2018.
- [16] H. J. A. a. S. Singh, "How Generative AI is Revolutionizing Scientific Research by Automating Hypothesis Generation," 2025.
- [17] Y. Gao *et al.*, "Retrieval-Augmented Generation for Large Language Models: A Survey," vol. abs/2312.10997, 2023.
- [18] A. J. I. J. o. F. E. I. Abisoye, "A Conceptual Framework for Integrating Artificial Intelligence into STEM Research Methodologies for Enhanced Innovation," 2024.
- [19] E. Chávez - Angel *et al.*, "Applied Artificial Intelligence in Materials Science and Material Design," p. 2400986, 2025.
- [20] C. Zuccarini, K. Ramachandran, and D. D. J. A. M. Jayaseelan, "Material discovery and modeling acceleration via machine learning," vol. 12, no. 9, p. 090601, 2024.
- [21] I. H. J. S. c. s. Sarker, "AI-based modeling: techniques, applications and research issues towards automation, intelligent and smart systems," vol. 3, no. 2, p. 158, 2022.

- [22] A. Checco, L. Bracciale, P. Loreti, S. Pinfield, G. J. H. Bianchi, and S. S. communications, "AI-assisted peer review," vol. 8, no. 1, pp. 1-11, 2021.
- [23] N. Fabiano *et al.*, "How to optimize the systematic review process using AI tools," vol. 4, no. 2, p. e12234, 2024.
- [24] J. de la Torre-López, A. Ramírez, and J. R. J. C. Romero, "Artificial intelligence to automate the systematic review of scientific literature," vol. 105, no. 10, pp. 2171-2194, 2023.
- [25] R. Whittemore, A. Chao, M. Jang, K. E. Minges, C. J. H. Park, and Lung, "Methods for knowledge synthesis: an overview," vol. 43, no. 5, pp. 453-461, 2014.
- [26] S. Aryal *et al.*, "Leveraging multi-AI agents for cross-domain knowledge discovery," 2024.
- [27] M. Fan, J. Hu, R. Cao, W. Ruan, and X. J. C. Wei, "A review on experimental design for pollutants removal in water treatment with the aid of artificial intelligence," vol. 200, pp. 330-343, 2018.
- [28] H. Mazaheri, M. Ghaedi, M. A. Azghandi, and A. J. P. C. C. P. Asfaram, "Application of machine/statistical learning, artificial intelligence and statistical experimental design for the modeling and optimization of methylene blue and Cd (II) removal from a binary aqueous solution by natural walnut carbon," vol. 19, no. 18, pp. 11299-11317, 2017.
- [29] W. Rodgers, S. Al-Shaikh, and M. J. I. T. o. E. M. Khalil, "Protocol analysis data collection technique implemented for artificial intelligence design," vol. 71, pp. 6842-6853, 2023.
- [30] R. Dahiya *et al.*, "AI Tools for Automating Data Collection and Analysis," in *Revolutionizing Academic Research With AI and Augmented Reality*: IGI Global Scientific Publishing, 2026, pp. 459-486.
- [31] A. Radetzky and A. J. A. i. i. m. Nürnberger, "Visualization and simulation techniques for surgical simulators using actual patient's data," vol. 26, no. 3, pp. 255-279, 2002.
- [32] B. Kovalerchuk, K. Nazemi, R. Andonie, N. Datia, and E. Banissi, *Integrating artificial intelligence and visualization for visual knowledge discovery*. Springer, 2022.
- [33] M. Khalifa, M. J. C. M. Albadawy, and P. i. B. Update, "Using artificial intelligence in academic writing and research: An essential productivity tool," vol. 5, p. 100145, 2024.
- [34] T. J. J. o. A. I. Said, "The Future of Academic Writing: How AI is Revolutionizing Citation Management," vol. 16, no. 2, pp. 520-539, 2025.
- [35] G. Wagner, R. Lukyanenko, and G. J. J. o. I. T. Paré, "Artificial intelligence and the conduct of literature reviews," vol. 37, no. 2, pp. 209-226, 2022.
- [36] E. Y. J. a. p. a. Chang, "Unlocking the Wisdom of Large Language Models: An Introduction to The Path to Artificial General Intelligence," 2024.
- [37] S. Ramalingam, *RAG in Action: Building the Future of AI-Driven Applications*. Libertatem Media Private Limited, 2023.
- [38] A. Salve, S. Attar, M. Deshmukh, S. Shivpuje, and A. M. J. A. Utsab, "A Collaborative Multi-Agent Approach to Retrieval-Augmented Generation Across Diverse Data," vol. abs/2412.05838, 2024.
- [39] E. J. I. J. o. A. I. Novák, Data Science, and M. Learning, "Automated Machine Learning (AutoML): Challenges and Future Trends in AI Model Optimization," vol. 1, no. 1, pp. 11-21, 2020.
- [40] X. Xu *et al.*, "A comprehensive review on synergy of multi-modal data and ai technologies in medical diagnosis," vol. 11, no. 3, p. 219, 2024.
- [41] A. Mohsin, H. Janicke, A. Wood, I. H. Sarker, L. Maglaras, and N. J. a. p. a. Janjua, "Can we trust large language models generated code? a framework for in-context learning, security patterns, and code evaluations across diverse llms," 2024.
- [42] N. S. Alias, M. J. Abd Rani, A. N. Che Kasim, F. Masnan, and N. Zainol Abidin, "Artificial Intelligence (AI) in Social Sciences," 2024.
- [43] R. Kusters *et al.*, "Interdisciplinary research in artificial intelligence: challenges and opportunities," vol. 3, p. 577974, 2020.
- [44] B. V. Calster *et al.*, "Performance evaluation of predictive AI models to support medical decisions: Overview and guidance," vol. abs/2412.10288, 2024.
- [45] G. S. Handelman *et al.*, "Peering Into the Black Box of Artificial Intelligence: Evaluation Metrics of Machine Learning Methods," vol. 212 1, pp. 38-43, 2019.
- [46] S. T. H. Mortaji, M. E. J. I. J. o. I. i. M. Sadeghi, Economics, and S. Sciences, "Assessing the Reliability of Artificial Intelligence Systems: Challenges, Metrics, and Future Directions," 2024.
- [47] Z. Chen *et al.*, "Research integrity in the era of artificial intelligence: Challenges and responses," vol. 103, 2024.
- [48] A. K. Guerrero-Solís, G. F. Ruiz-Muñoz, D. A. Yépez-González, and M. N. J. M. L. A. J. Sánchez-Lascano, "El impacto de la inteligencia artificial en la producción científica," 2025.
- [49] R. Qureshi *et al.*, "AI in drug discovery and its clinical relevance," vol. 9, no. 7, 2023.
- [50] P. Tomczyk, P. Brüggemann, and D. J. E. J. o. B. Vrontis, "AI meets academia: transforming systematic literature reviews," 2024.
- [51] U. C. AJUZIEOGU, "Towards Hallucination-Resilient AI: Navigating Challenges, Ethical Dilemmas, and Mitigation Strategies,"
- [52] B. A. Malin, K. E. Emam, and C. M. J. J. o. t. A. m. i. a. O'Keefe, "Biomedical data privacy: problems, perspectives, and recent advances," vol. 20, no. 1, pp. 2-6, 2013.
- [53] M. A. Haque *et al.*, "SOK: Exploring Hallucinations and Security Risks in AI-Assisted Software Development with Insights for LLM Deployment," vol. abs/2502.18468, 2025.
- [54] W. Leong and J. J. A. S. J. Zhang, "AI on academic integrity and plagiarism detection," vol. 20, p. 75, 2025.
- [55] N. B. Ganesh, D. Siddineni, V. V. Reddy, K. Lateef, R. J. J. o. I. S. E. Sharma, and Management, "Corporate Governance in the Age of AI: Ethical Oversight and Accountability Frameworks," 2025.
- [56] M. Mitchell *et al.*, "Model cards for model reporting," in *Proceedings of the conference on fairness, accountability, and transparency*, 2019, pp. 220-229.
- [57] D. Brajovic *et al.*, "Model Reporting for Certifiable AI: A Proposal from Merging EU Regulation into AI Development," vol. abs/2307.11525, 2023.
- [58] M. Gridach, J. Nanavati, K. Z. E. Abidine, L. Mendes, and C. D. J. A. Mack, "Agentic AI for Scientific Discovery: A Survey of Progress, Challenges, and Future Directions," vol. abs/2503.08979, 2025.
- [59] H. Dawid, P. Harting, H. Wang, Z. Wang, and J. Yi, "Agentic Workflows for Economic Research: Design and Implementation," 2025.
- [60] A. J. J. o. C. S. Satav and T. Studies, "Enterprise API & Platform Strategy in the era of Agentic AI," 2025.
- [61] L. Hughes *et al.*, "AI Agents and Agentic Systems: A Multi-Expert Analysis," vol. 65, pp. 489 - 517, 2025.